



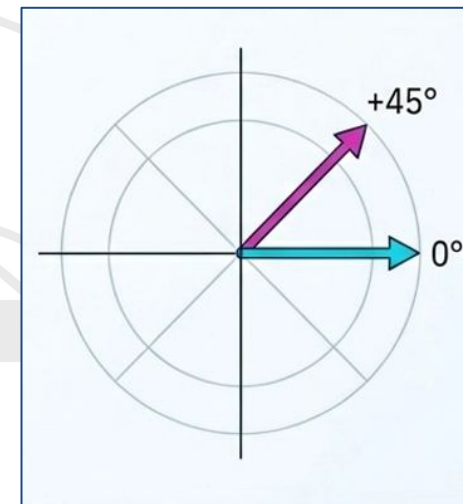
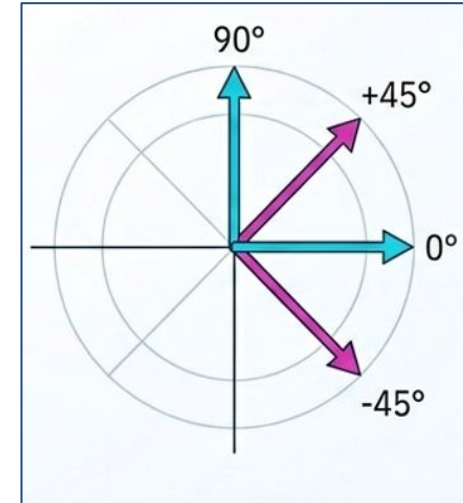
Quantum Key Distribution:

B92 vs. SIM BPSK



Introduction to the B92 Protocol

- It was proposed by Charles Bennett in 1992.
- Evolution from **four-state protocols** (like in BB84) to a minimalist **two-state approach**.
- The shift to B92 simplifies the QKD mechanism while maintaining **robust security** through non-orthogonal states.
- Information is encoded into single photon using 2 non-orthogonal distinct polarization states.
- Sifting relies on Bob's **announcement of conclusive or inconclusive detections** regardless of his **basis**.
- Since the two states are non-orthogonal, **no single quantum measurement can perfectly discriminate between them**.
- Eve cannot deterministically read the states without causing detectable errors.



Execution Steps of B92

1. State Preparation

Alice generates a bit stream (0 or 1) and chooses one of the states out of a pair of non-orthogonal states (like 0° and 45°) to encode the bit into a photon's polarization state.

2. Random Measurement

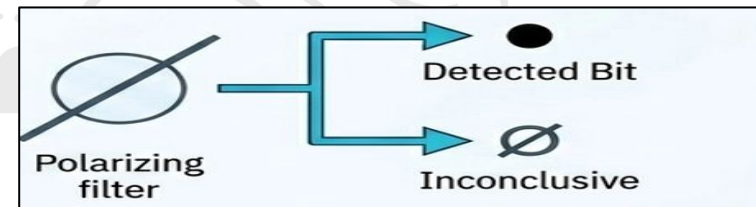
Bob receives the photons but doesn't know Alice's photon state. He randomly selects a polarisation filter (90° or -45°) for each incoming photon and records the result.

3. Sifting

Over a classical public channel, Bob announces whether he has **conclusive or detected** bits or **inconclusive or discarded** bits. He discards where he has inconclusive or no detection.

Bob is not looking for what Alice sent; he is looking for what Alice could not have sent.

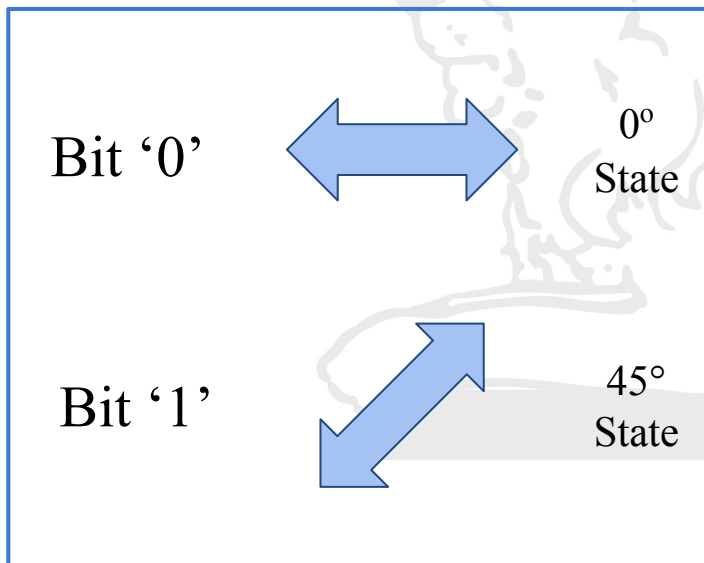
Bit 0 → State 0°
 Bit 1 → State $+45^\circ$



Execution Steps of B92

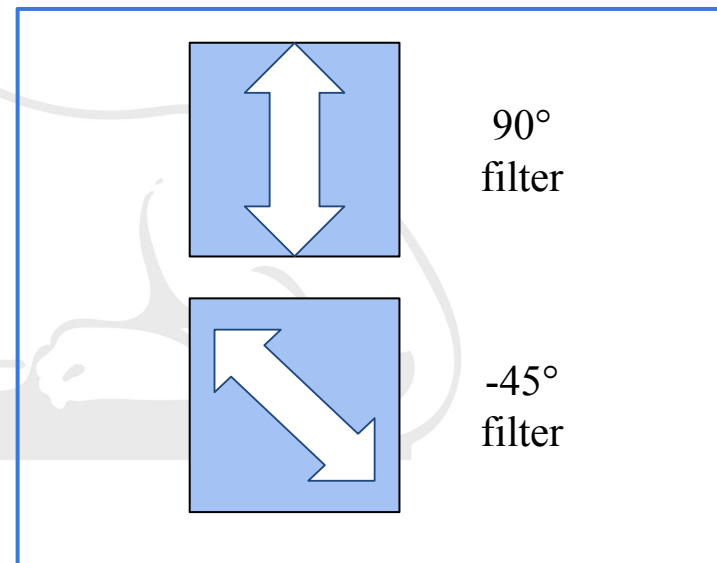
Alice State Preparation Mechanics

- Alice generates a sequence of bits.
- Bits are mapped to non-orthogonal photon polarization quantum states.



Bob State Measurement

- Bob passes the incoming photons through its polarization filter.
- Bob check whether the polarization filter passes something or not.



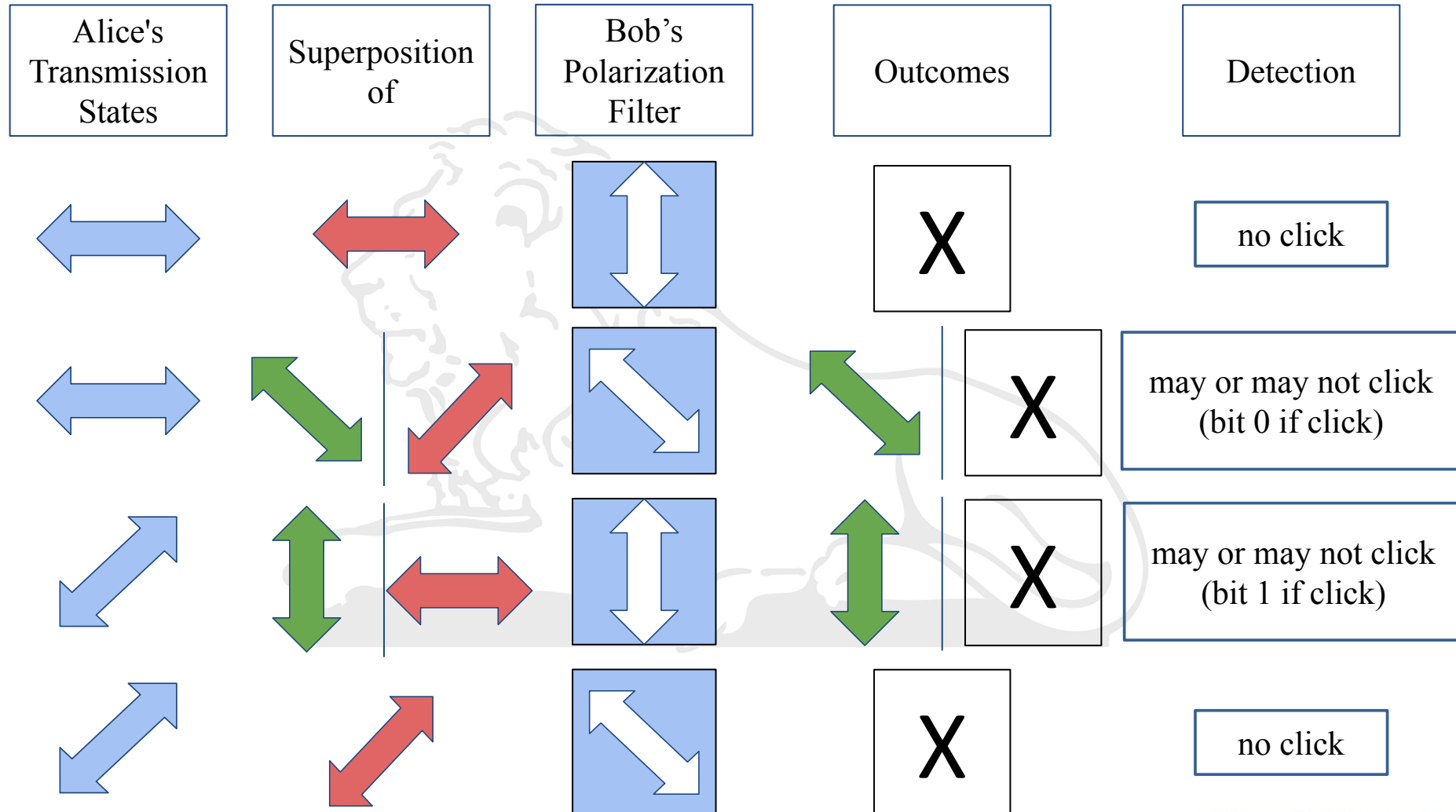
Execution Steps of B92

$$|0^\circ\rangle = \frac{1}{\sqrt{2}}(|45^\circ\rangle + |-45^\circ\rangle)$$

$$|45^\circ\rangle = \frac{1}{\sqrt{2}}(|0^\circ\rangle + |90^\circ\rangle)$$



Bob's measurement possibilities



Execution Steps of B92

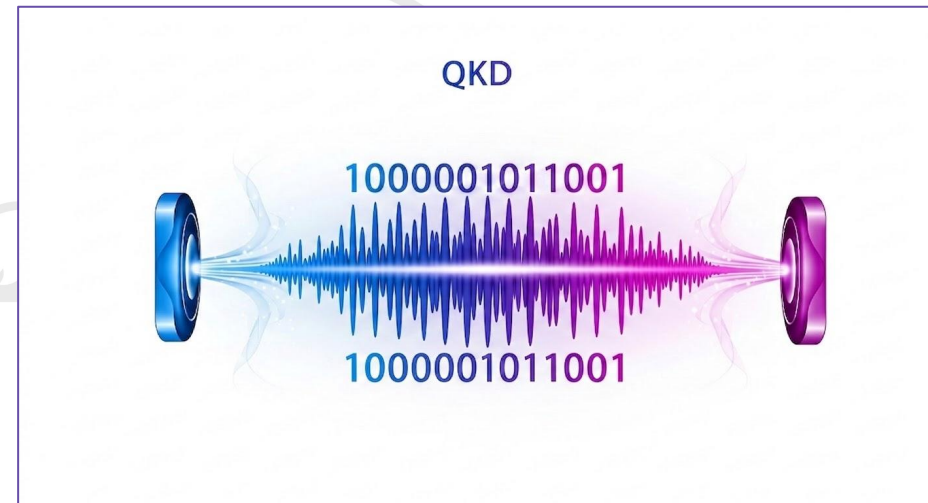
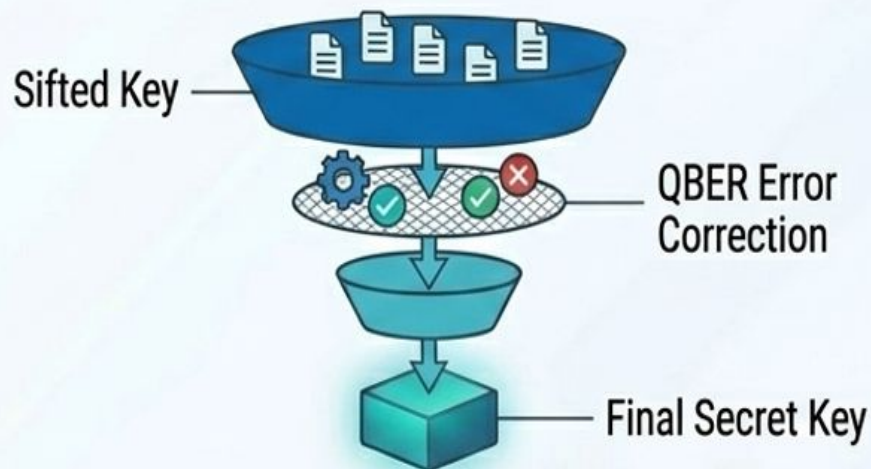
Example of B92 transmission

Bit Transmitted	1	0	1	0	1	0	1	1	1	0	0	1	0	0
Alice Transmission State	45	0	45	0	45	0	45	45	45	0	0	45	0	0
Bob Measurement State	90	-45	-45	-45	90	90	-45	90	-45	90	90	-45	-45	90
Measurement output	X or 1	X or 0	X	X or 0	X or 1	X	X	X or 1	X	X	X	X	X or 0	X
Possible Bit Detected	1	X	X	0	X	X	X	1	X	X	X	X	0	X

Execution Steps of B92

Parameter Estimation

- Bob publicly discloses the temporal indices of conclusive or inconclusive detection events to Alice, without revealing bit values.
- Alice synchronizes her transmission logs with Bob's conclusive time instances to extract the raw sifted key from the data set.
- A random subset of the sifted key with indices is exchanged over a public channel to empirically estimate the Quantum Bit Error Rate (QBER).
- If errors exceed threshold, abort (Eve is present). Otherwise, correct errors and apply post-processing.



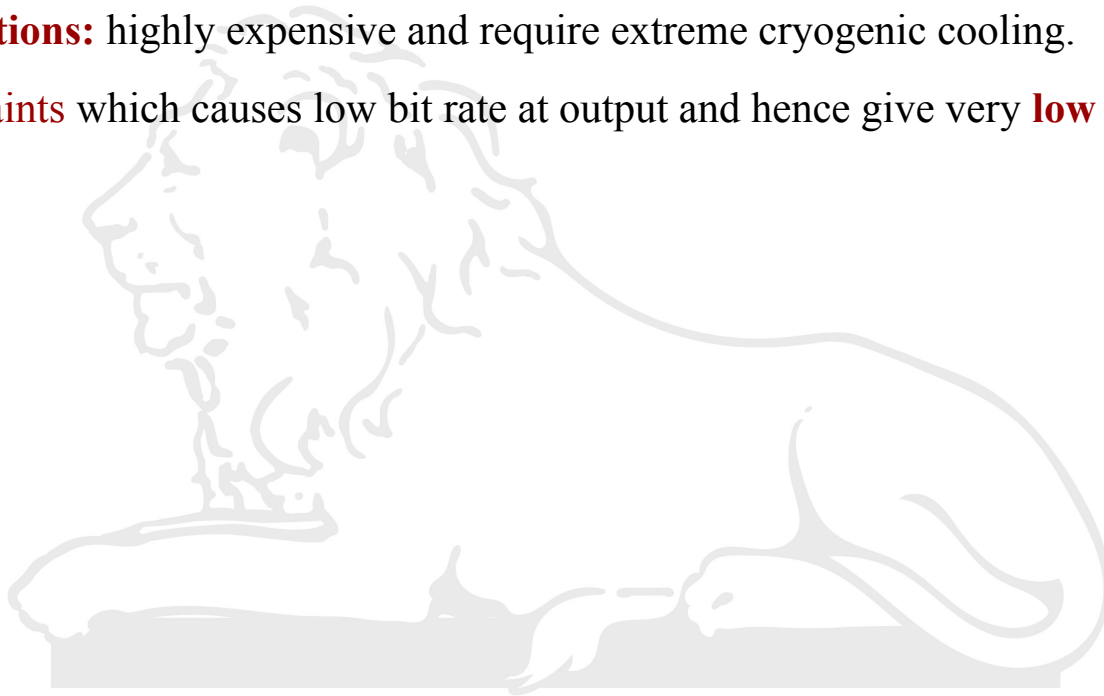
Simulation of B92

https://www.st-andrews.ac.uk/physics/quvis/simulations_html5/sims/cryptography-b92/B92_photons.html



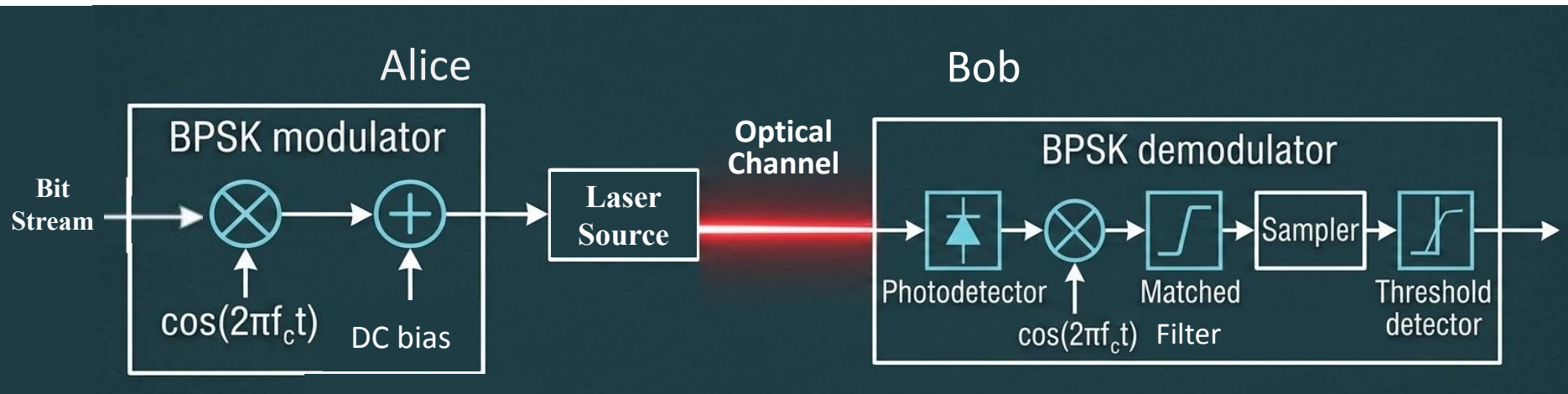
Limitations of B92

- B92 is DV-QKD protocol which is **not compatible** with current telecom equipments.
- **Source limitations:** true single photon sources are highly complex and expensive
- **Detector limitations:** highly expensive and require extreme cryogenic cooling.
- **Distance constraints** which causes low bit rate at output and hence give very **low key generation rate**.



Subcarrier Intensity Modulation Direct Detection

- SIM-DD can be implemented using conventional optical architecture with direct modulation.
- Employs a continuous-wave (CW) laser source as an optical carrier, which is modulated by a radio-frequency (RF) subcarrier signal.
- Instead of on-off keying, Bits '0' and '1' are mapped to the RF-subcarrier, characterized by a phase shift of 180° .



SIM-BPSK Block Diagram

Gaussian CV-QKD to DM-CV-QKD

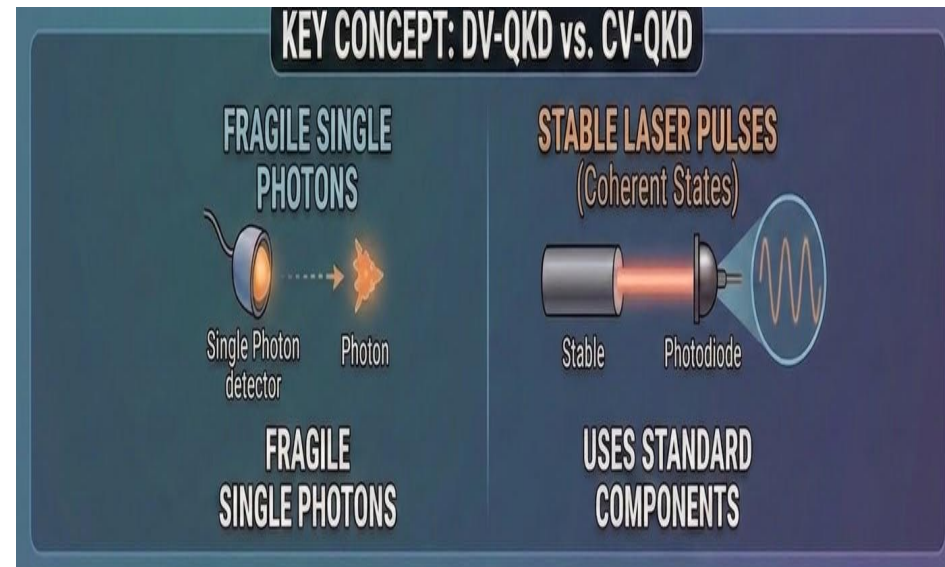
Transition to Quantum: Gaussian CV-QKD to DM-CV-QKD

- **Gaussian Continuous-Variable QKD**

(CV-QKD): Encodes quantum information into the amplitude and phase of a laser field, and enabling the use of standard telecom components.

- **Discrete-Modulation (DM-CV-QKD):**

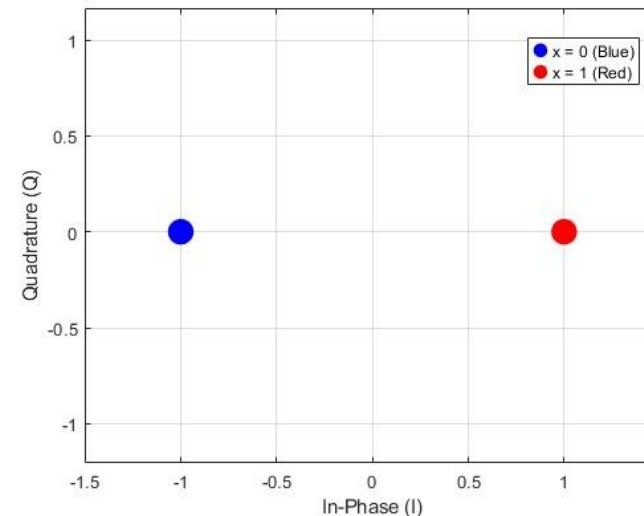
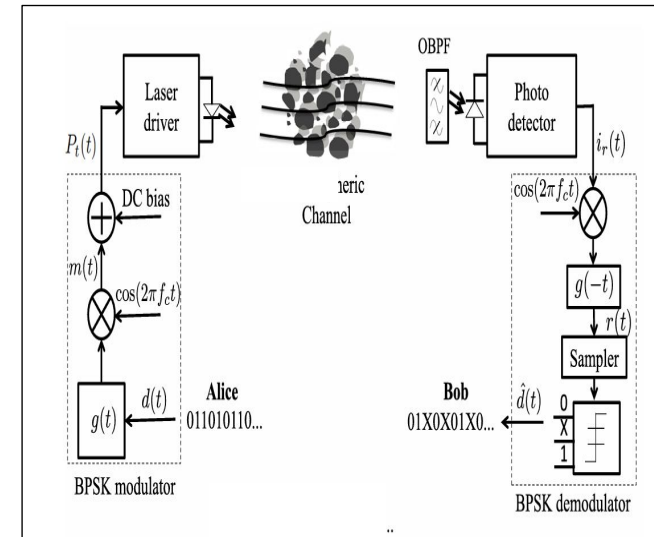
Simplifies processing by restricting these continuous signals to a finite set of states (e.g., a two-point constellation).



CV-QKD to DM-CV-QKD

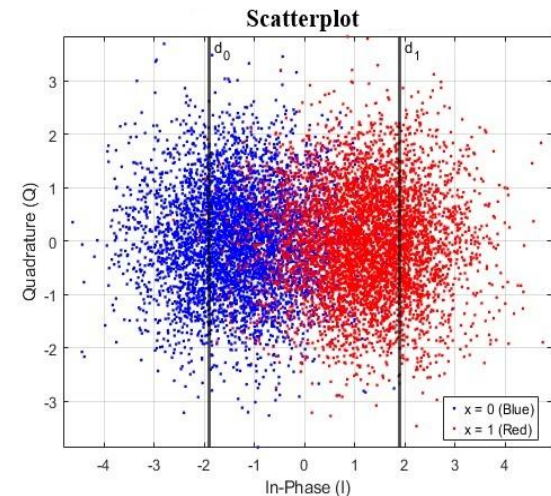
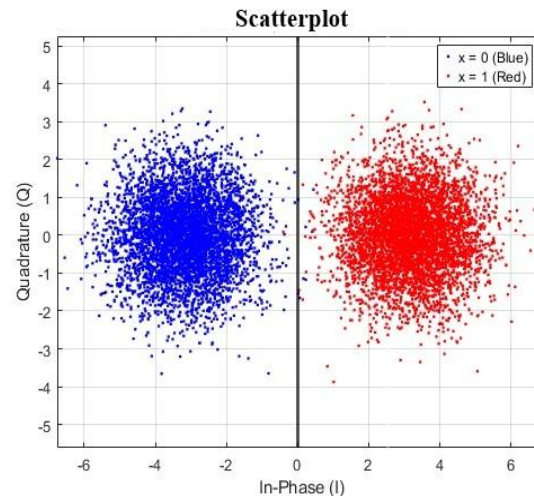
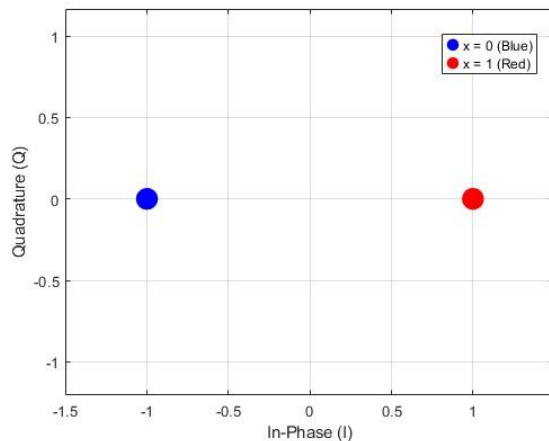
SIM-BPSK in DM-CV-QKD

- The SIM-BPSK Connection:** Acts as a practical DM-CV-QKD implementation. It uses an RF subcarrier to map phase states into modulated intensity levels, allowing for simple Direct Detection (DD).
- B92 Protocol Equivalence:** By encoding the key into two different non orthogonal phases ('0°' and '180°'), this system serves directly as the continuous-variable equivalent of the classic DV-QKD B92 protocol.



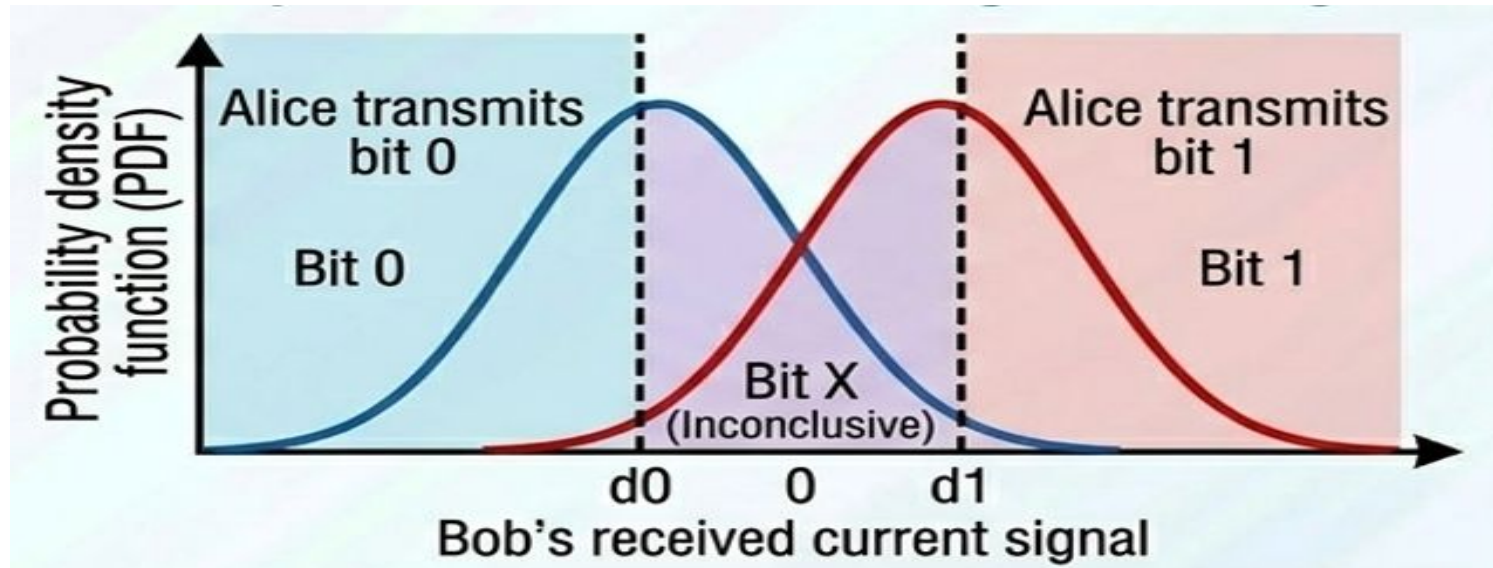
SIM BPSK (DM-CV-QKD)

- A very small modulation depth (δ) is intentionally chosen to ensure the '0' and '1' quantum states strongly overlap.
- **Constellation Points:** This creates two distinct points on the real axis of the constellation diagram of the transmitter, representing bits '0' and '1'.
- **Modulation depth (δ):** Used to bring the constellation 'clouds' closer together, creating the overlap needed for quantum security.
- **Noise Impact:** Noise (shot noise, excess noise, etc.) causes constellation points to 'spread' into clouds, their overlap in the systems leads to decision errors.
- **Dual-Threshold (ρ):** Instead of using a single threshold, Bob uses two thresholds, creating an erasure zone between states.



SIM BPSK (DM-CV-QKD)

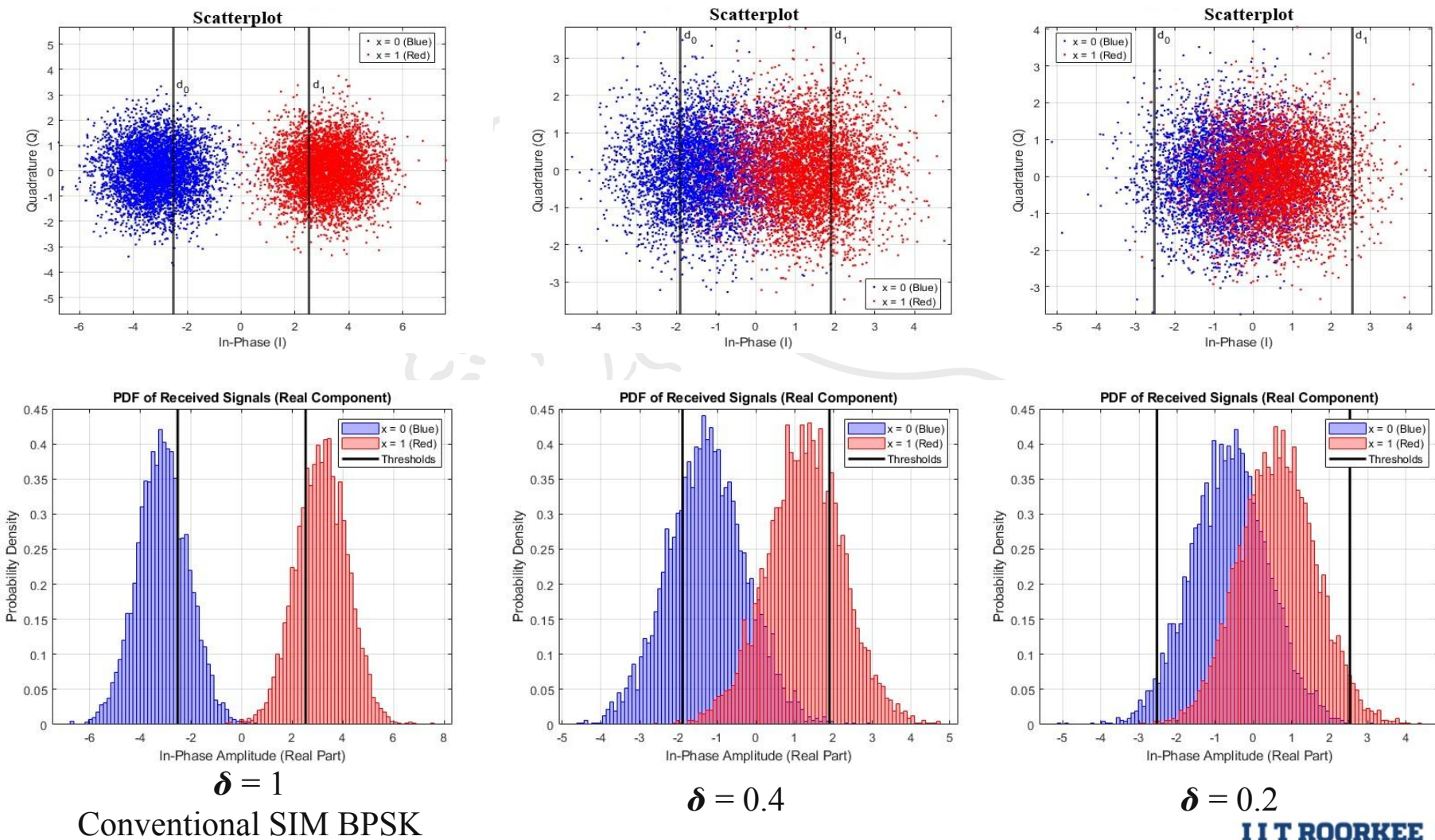
Detection mechanism



Dual-Threshold Direct Detection creates artificial conclusive and inconclusive bins, perfectly mimicking the non-orthogonal sifting process of B92.

SIM BPSK (DM-CV-QKD)

Effect of modulation depth and threshold



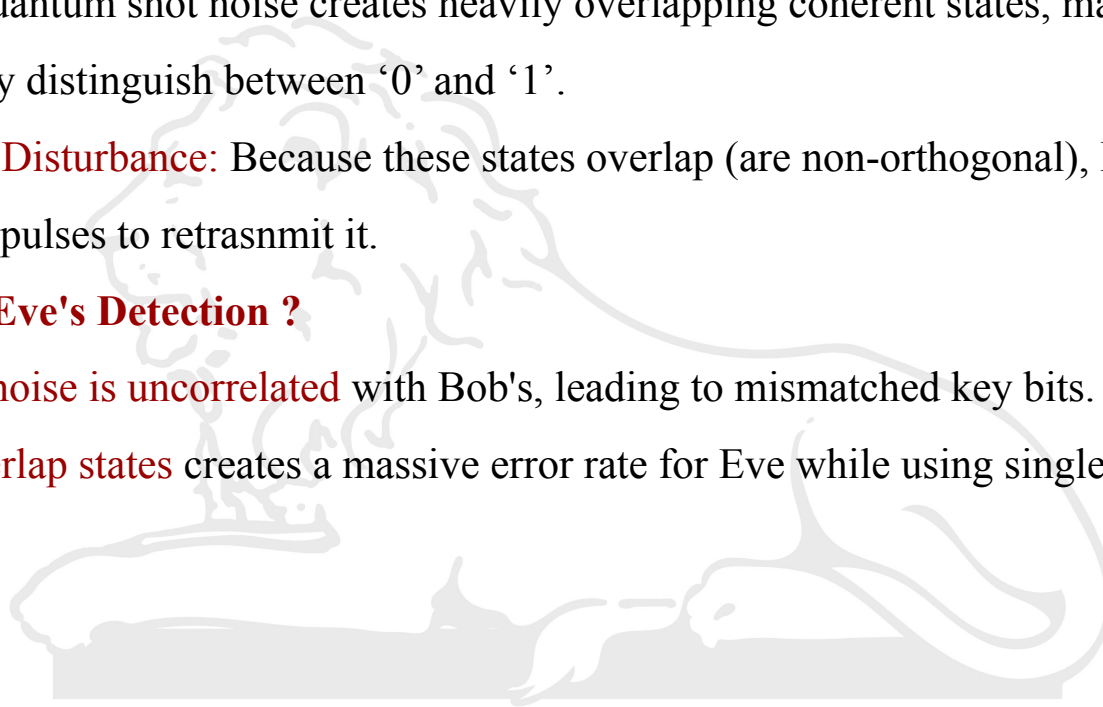
SIM BPSK (DM-CV-QKD)

Security Mechanisms in SIM-BPSK QKD

- **Indistinguishable States (Heisenberg Uncertainty):** Low modulation depth (δ) combined with fundamental quantum shot noise creates heavily overlapping coherent states, making it impossible for Eve to perfectly distinguish between '0' and '1'.
- **No-Cloning & Disturbance:** Because these states overlap (are non-orthogonal), Eve cannot perfectly clone the laser pulses to retransmit it.

Why high error in Eve's Detection ?

- Eve's channel **noise is uncorrelated** with Bob's, leading to mismatched key bits.
- The **highly overlap states** creates a massive error rate for Eve while using single threshold.



B92 vs SIM BPSK

Protocol Feature	B92 (DV-QKD)	SIM-BPSK (DM-CV-QKD)
Information Carrier	Single Photon (Discrete Variables)	Optical Intensity (Continuous Variables)
Encoding Mechanism	Polarization States (2 non-orthogonal states)	Amplitude / Intensity Variation (δ) (180° separation)
Indistinguishability Reason	Non-orthogonal overlapping	Small modulation depth
Discard Trigger	Wrong selection of polarization filter..	Signal lies in discarded region
Eavesdropper Barrier	No-cloning theorem / State collapse	Heisenberg Uncertainty, No-cloning theorem & Disturbance
Hardware Complexity	Extremely High (Cryogenics, low dark-count SPDs)	Relatively Low (Standard telecom components)

MATLAB Implementation SIM BPSK

Some essentials MATLAB commands

- **clc** – Clears the command window (removes clutter but keeps your variables).
- **clear** or **clear all** – Removes all variables from the workspace (erases memory).
- **close all** – Closes all open figure/plot windows.
- **disp(x)** – Displays the contents of variable x or a text string without printing the variable name.
- **zeros(m, n)** – Creates an m by n matrix filled with zeros.
- **ones(m, n)** – Creates an m by n matrix filled with ones.
- **rand(m, n)** – Creates m by n matrix of uniformly distributed random numbers between 0 and 1.
- **linspace(x1, x2, n)** – Generates n evenly spaced points between x1 and x2. 3.
- **sum(x)** – Adds up all the elements in array x.
- **sqrt(x)** – Calculates the square root.
- **length(x)** – Returns the length of an array.
- **plot(x, y)** – Creates a 2D line plot of y versus x.
- **figure** – Opens a new, blank figure window so you don't overwrite your last plot.
- **xlabel('Text')** and **ylabel('Text')** – Adds labels to your X and Y axes.
- **title('Text')** – Adds a title to the top of your plot.
- **hold on** – Retains the current plot so you can add another line/graph on top of it.

MATLAB Implementation SIM BPSK

Alice's Modulation Protocol

Signal Generation

Alice generates an array of random bits and assigns distinct amplitudes. Bit 1 receives a higher amplitude, while Bit 0 receives a proportionally lower one based on the modulation depth parameter.

$$I_0 = -\delta \text{ and } I_1 = \delta$$

Bob's Dual-Threshold Strategy

Filtering Uncertainties

Unlike standard detection, Bob employs two distinct boundaries. Signals exceeding the upper limit become Bit 1, while those falling below the lower limit become Bit 0. Ambiguous signals in the middle are entirely discarded.

$$X_b = \sqrt{TX_a} + N_a \text{ where } N_a \text{ is total noise } (0, \sigma^2)$$

$$\text{and } X_a \in (I_0, I_1)$$

$$d_0 = -\varrho\sqrt{T} \text{ and } d_1 = \varrho\sqrt{T}$$

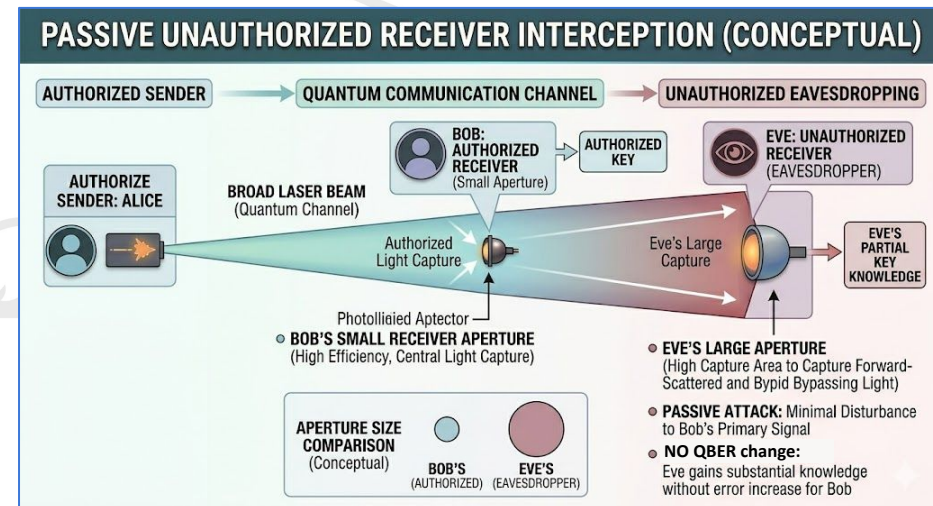
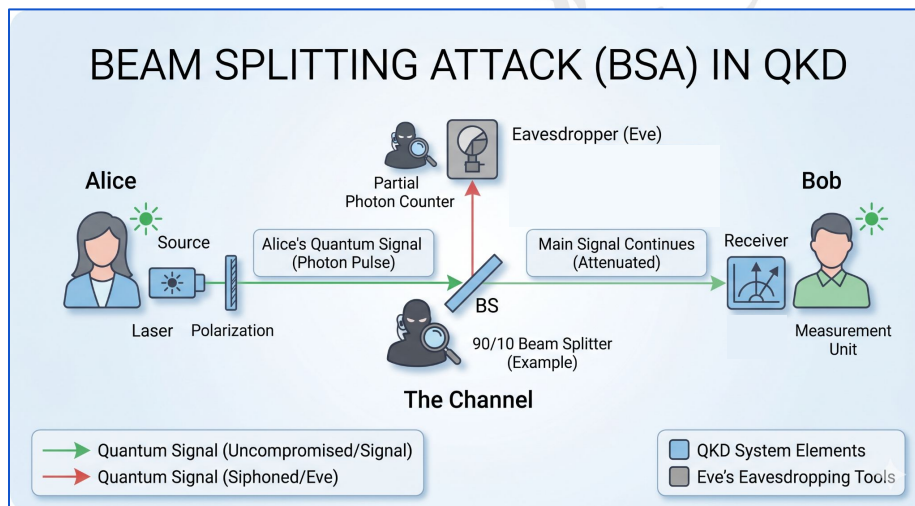
MATLAB Implementation SIM BPSK

Eve's Interception

- The eavesdropper (Eve) uses a standard, **single-threshold direct detection** strategy, with the decision boundary fixed at the origin or conventional zero thresholding.
- This single threshold, necessary to capture all signals, forces Eve into a **detection region with a high error rate** near the base signal amplitude, causing additional detection errors.

$$d_e = 0$$

- This causes BER of Eve **always higher than** QBER of bob.



MATLAB Implementation SIM BPSK

Performance Metrics

Sifting Probability

The fraction of bits Bob confidently detects and accepts over the total transmitted bits.

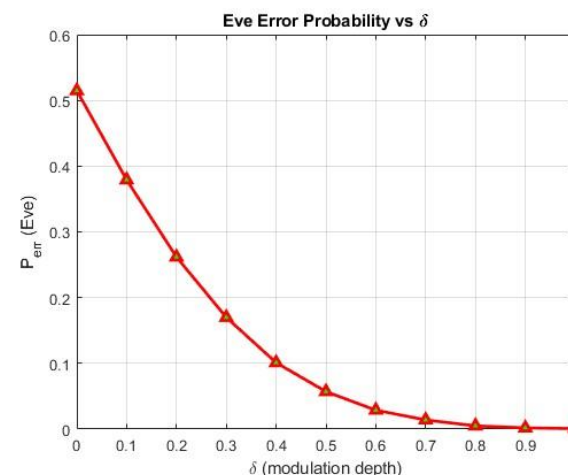
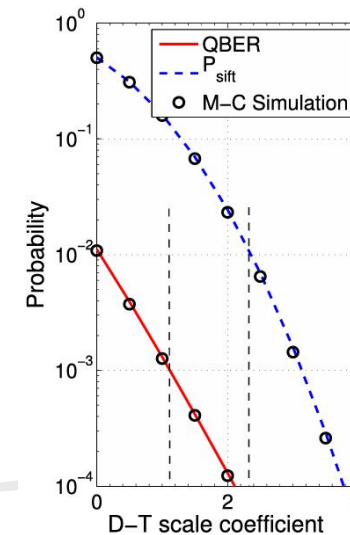
Higher thresholds mean lower sifting rates.

QBER

Quantum Bit Error Rate representing the ratio of mismatched bits between Alice and Bob's sifted sets. The ultimate measure of channel fidelity.

Eve's Error Rate

The baseline error probability experienced by the eavesdropper due to channel noise, without the benefit of post-selection sifting.



Thank you